

Digital Electronics (MCQ મુજબ સંપૂર્ણ ટોપિક સમજણ)

સરળ ગુજરાતી નોટ્સ – Step by Step

1. Number System (નંબર સિસ્ટમ)

Number System શું છે?

કમ્પ્યુટર અને ડિજિટલ સર્કિટમાં નંબર દર્શાવવાની પદ્ધતિ.

પ્રકાર	Base	Digits
Binary	2	0,1
Octal	8	0-7
Decimal	10	0-9
Hexadecimal	16	0-9, A-F

Binary Number System

ડિજિટલ સર્કિટ માટે સૌથી મહત્વપૂર્ણ.

ઉદાહરણ:

$$(1011)_2$$

Decimal માં:

$$1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$
$$= 8 + 0 + 2 + 1 = 11$$

Hexadecimal System

Hex	Binary
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

મહત્વપૂર્ણ Shortcut

Decimal	Binary
1	0001
2	0010
4	0100
8	1000

Decimal

16

Binary

10000

2. Boolean Algebra

Boolean Algebra માં માત્ર બે value હોય:

- 0 = LOW
- 1 = HIGH

મુખ્ય Operations

Symbol

Meaning

+

OR

.

AND

'

NOT

Boolean Laws

Identity Law

$$A + 0 = A$$

$$A \cdot 1 = A$$

Complement Law

$$A + A' = 1$$

$$A \cdot A' = 0$$

Idempotent Law

$$A + A = A$$

$$A \cdot A = A$$

Absorption Law

$$A + AB = A$$

De Morgan's Theorem

$$(A \cdot B)' = A' + B'$$

અને

$$(A + B)' = A' \cdot B'$$

3. Logic Gates

AND Gate

બધા input 1 હોય ત્યારે output 1

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

Equation:

$$Y = A \cdot B$$

OR Gate

એક input પણ 1 હોય તો output 1

Equation:

$$Y = A + B$$

NOT Gate

Input નો opposite આપે.

Equation:

$$Y = A'$$

XOR Gate

Input અલગ હોય ત્યારે HIGH

Equation:

$$Y = A'B + AB'$$

Shortcut:

- Same = 0
- Different = 1

XNOR Gate

Equality detector.

- Same = 1
- Different = 0

Universal Gates

Gate	Reason
NAND	કોઈપણ logic બનાવી શકાય
NOR	કોઈપણ logic બનાવી શકાય

4. Karnaugh Map (K-Map)

Boolean expression simplify કરવા માટે.

K-map Grouping Rules

Group	Variables Remove
2	1 variable
4	2 variables
8	3 variables

Important Points

- Group power of 2 માં જ થાય
- Maximum grouping કરો
- Don't care simplify માટે ઉપયોગી

5. Adders

Half Adder

Inputs:

- A
- B

Outputs:

- Sum
- Carry

Half Adder Equations

Sum:

$$S = A \oplus B$$

Carry:

$$C = A \cdot B$$

Full Adder

Inputs:

- A
- B
- Carry in

Outputs:

- Sum
- Carry out

6. Subtractor

Half Subtractor

Difference:

$$D = A \oplus B$$

Borrow:

$$B = A' \cdot B$$

7. Complements

1's Complement

All bits reverse.

Example:

1010 $\rightarrow$ 0101

2's Complement

1's complement + 1

Example:

1010

↓

0101

↓

0110

8. Codes

BCD Code

Decimal digits $\rightarrow$ binary representation.

Example:

9 = 1001

Excess-3 Code

Decimal + 3

Example:

— — —

$$4 + 3 = 7$$

$$7 = 0111$$

Gray Code

Adjacent values માં માત્ર 1 bit change થાય.

Conversion માટે XOR ઉપયોગ થાય.

9. Multiplexer (MUX)

Multiplexer શું કરે?

ઘણા inputs માંથી એક select કરે.

તેને Data Selector કહે.

4:1 MUX

Inputs

4

Select Lines

2

Formula:

$$2^n = Inputs$$

10. Demultiplexer

એક input ને ઘણા outputs માં મોકલે.

તેને Data Distributor કહે.

11. Decoder

Binary input ને one-hot output માં convert કરે.

Example

3-to-8 Decoder:

- 3 inputs
- 8 outputs

Formula:

$$Outputs = 2^n$$

12. Encoder

ધણ inputs ને binary output માં convert કરે.

Priority Encoder

એકથી વધારે inputs HIGH હોય તો priority પ્રમાણે output આપે.

13. ROM / PLA / PAL

ROM

Read Only Memory

Permanent data store કરે.

PLA

Programmable AND + OR arrays

Flexible

PAL

Programmable AND array only

Fast

14. Sequential Circuits

Difference

Combinational

Memory નથી

Present input પર depend

Sequential

Memory હોય

Previous state પર depend

15. Flip-Flops

Flip-Flop શું છે?

1 bit data store કરતું memory element.

SR Flip-Flop

S	R	Output
0	0	No change
0	1	Reset
1	0	Set
1	1	Invalid

D Flip-Flop

Input જે હોય તે output બને.

Equation:

$$Q = D$$

$$Q^+ = D$$

T Flip-Flop

T=1 હોય તો toggle થાય.

Equation:

$$Q^+ = T \oplus Q$$

JK Flip-Flop

Universal Flip-Flop

Race around problem આવે.

Equation:

$$Q^+ = JQ' + K'Q$$

Race Around Condition

JK flip-flop માં continuous toggling.

Solution:

- Master Slave FF
- Edge Triggering

16. Registers

Register શું છે?

ઘણા flip-flops નું group.

Data store કરે.

Shift Registers

Type	Meaning
SISO	Serial In Serial Out
SIPO	Serial In Parallel Out
PISO	Parallel In Serial Out
PIPO	Parallel In Parallel Out

Left Shift

$\times 2$

Right Shift

$\div 2$

17. Counters

Counter શું છે?

Clock pulses count કરે.

Ripple Counter

Asynchronous counter.

Slow because propagation delay.

Synchronous Counter

બધા FF ને same clock.

Fast.

Modulus (MOD)

Counter ÷ total states.

Example:

MOD-10 = 10 states

Ring Counter

One-hot code use કરે.

4-bit ring counter → 4 states

Johnson Counter

Twisted Ring Counter

4-bit Johnson → 8 states

18. State Machines

Moore Machine

Output depends only on state.

Mealy Machine

Output depends on:

- Input
- Present state

Moore vs Mealy

Moore

Stable output

More states

Mealy

Fast response

Fewer states

Important Exam Shortcuts

Topic

XOR

XNOR

NAND

NOR

Right Shift

Left Shift

D FF

JK FF

Ripple Counter

Synchronous Counter

Shortcut

Different = 1

Same = 1

Universal gate

Universal gate

$\div 2$

$\times 2$

Store data

Universal FF

Slow

Fast

this data ok as it but in add more deatils in sort like
explation and used ya impotred add rcetra this details and
make one pdf in add all of that date in perfect formate in
give me that pdf

pdf in not show perfect make perfect formate in look like
organl pdf make

Stopped thinking >

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